

Edge-AI GEMM Accelerator: 8x8 Int8/Int32 Systolic Array

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Project Repository: <https://github.com/Saswatpadhy9905/8x8-INT8-Systolic-Array-GEMM-Accelerator>

Abstract

This report outlines the microarchitecture, verification, and physical implementation of a high-performance General Matrix Multiply (GEMM) accelerator, featuring an 8x8 output-stationary systolic array tailored for edge-AI inference. Implemented in SystemVerilog for the Xilinx Artix-7 (XC7A100T) FPGA, it achieves a compute throughput of 13.3 GOPS (6.65 GMAC/s). The design optimally maps parallel processing elements (PEs) to dedicated DSP48E1 slices, minimizing fabric routing congestion and achieving physical timing closure at 104 MHz with a Worst Negative Slack (WNS) of 0.41 ns.

Introduction

Matrix multiplication represents the dominant computational bottleneck in modern neural network inference. Systolic arrays mitigate this by maximizing data reuse and minimizing memory bandwidth requirements. This project implements a cycle-accurate 8x8 systolic array specifically designed for 8-bit integer (Int8) activations and weights, utilizing 32-bit (Int32) accumulators to prevent overflow during partial sum generation.

Hardware Architecture

The core architecture is an output-stationary 2D systolic grid comprising 64 parallel Processing Elements (PEs).

Processing Element (PE) Design

Each PE performs a Multiply-Accumulate (MAC) operation at every clock cycle. The inputs (Int8) are buffered and multiplied, while the stationary 32-bit accumulator continuously updates the partial sums. Once the matrix multiplication wavefront completes, the accumulated Int32 results are sequentially drained through the output buffers.

DSP Optimization and Routing

To achieve deterministic high-speed performance, the synthesis process was confined to route 100% of MAC operations into FPGA’s dedicated arithmetic blocks. The 64 PEs employ 64 DSP48E1 slices, utilizing native cascade routing to minimize routing congestion and lower dynamic power use.

Verification Methodology

The RTL was verified using a modern software-driven co-simulation framework.

- **Translation Engine:** Verilator was used to compile the SystemVerilog RTL into a high-speed C++ model.
- **Golden Model:** A structurally equivalent software model was developed in Python utilizing NumPy.
- **Regression Testing:** The C++ testbench drove over 10,000 randomized Int8 matrix tensors through both the RTL and the golden model, automatically asserting cycle-level functional equivalence with zero mismatches.

Implementation and Performance

The accelerator was synthesized and implemented Out-Of-Context (OOC) on the Xilinx Artix-7 (XC7A100T-csg324) using Vivado. Strategic pipeline insertion for high-fanout control nets and boundary skew buffering enabled full timing closure.

Table 1: Physical Implementation Metrics (Artix-7)

Metric	Value
Target Device	XC7A100T-csg324
Operating Frequency	104.37 MHz
Worst Negative Slack (WNS)	+0.41 ns
Compute Throughput	13.3 GOPS (6.65 GMAC/s)
DSP48E1 Utilization	64 / 64 (Forced Inference)
Slice LUTs	15,313
Slice Registers	35,464

Conclusion

The designed Edge-AI GEMM accelerator successfully demonstrates deterministic, high-throughput matrix multiplication suitable for edge inference environments. By combining cycle-accurate SystemVerilog RTL, robust C++/Verilator functional verification, and physical timing closure bounded by structural DSP mapping, the project highlights a complete end-to-end ASIC/FPGA digital design flow.